Topology Optimization 2

Gradients using adjoint sensitivity analysis:

1. Modify the code from last week to ***not*** use pyadjoint to calculate the gradients of the cost function and the volume constraint. Calculate instead the gradients using you own adjoint code. This includes both the Helmholtz filter and the linear elastic problem.
   Test your gradients using:
   a. Finite differences.
   b. Taylor series test.
   c. Compare with pyadjoint's gradients by simply printing some gradient component values.
2. Compare the running time of a complete optimization with your own gradients to an optimization with pyadjoint gradients. Can you beat pyadjoint?

Linear vs non-linear topology optimization of a cantilever beam:

3. Implement topology optimization for a linear elastic cantilever beam.
4. Implement topology optimization for a geometrically non-linear cantilever beam using a linear material model and Green-Lagrange strain tensor.
5. Compare the designs obtained using linear and non-linear models for small, medium, and large displacements.

Note: It can be difficult to solve non-linear problems and sometimes some tricks must be used if you have really large loads or deformations:

a) Gradually increasing the load in the non-linear mechanics solver is the most often used trick.
b) A non-linear material model can sometimes help with the behavior of low-density elements, see
   Towards the stabilization of the low density elements in topology optimization with large deformation, Ricardo Doll Lahuerta, Eduardo T. Simões, Eduardo M. B. Campello, Paulo M. Pimenta & Emilio C. N. Silva, Comput Mech (2013) 52:779–797, https://link.springer.com/article/10.1007/s00466-013-0843-x
c) If you don't want to bother with a or b, decrease your maximum load 😊